- Suvda kam eriydi, organik erituvchilarda yaxshi eriydi. Atsetilen suvda etilenga qaraganda yaxshiroq, atsetonda esa juda yaxshi eriydi.
- Atsetilen – rangsiz, deyarli hidsiz gaz, havodan biroz yengil (M = 26 g/mol). Kalsiy karbiddan olingan texnik atsetilen aralashmalar (PH_3, H_2S) hisobiga yoqimsiz hidli bo‘ladi.

Modda	Formula	Holati*	t_q, °C
atsetilen	C_2H_2	gaz	–84
propin	C_3H_4	gaz	–23
butin-1	C_4H_6	gaz	8
butin-2	C_4H_6	suyuq	27
pentin-1	C_5H_8	suyuq	40
pentin-2	C_5H_8	suyuq	56
geksin-1	C_6H_{10}	suyuq	71

* xona haroratida (20–25 °C)

4. Olinishi

1) Kalsiy karbidga suv ta’sir ettirish (karbid usuli)

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2\uparrow$$

Kalsiy karbid so‘ndirilmagan ohak bilan koksni elektr pechida (≈ 2000 °C) qizdirib olinadi: $CaO + 3C \rightarrow CaC_2 + CO$. Atsetilen laboratoriyada ham, sanoatda ham shu usulda olinadi. Magniy karbiddan esa propin hosil bo‘ladi:

$$Mg_2C_3 + 4H_2O \longrightarrow 2Mg(OH)_2 + CH_3{-}C{\equiv}CH$$

2) Metanni piroliz qilish (sanoat usuli)

$$2CH_4 \xrightarrow{1500\ °C} C_2H_2 + 3H_2$$

Reaksiya juda qisqa vaqt davomida o‘tkaziladi va hosil bo‘lgan gazlar tez sovitiladi, aks holda atsetilen uglerod va vodorodga parchalanib ketadi.

3) Uglerod va vodoroddan sintez (M. Bertlo, 1862-yil)

$$2C + H_2 \xrightarrow{\text{elektr yoyi}} C_2H_2$$

Elektr (volt) yoyi orasidan vodorod o‘tkazilganda atsetilen hosil bo‘ladi. Atsetilen oddiy moddalardan birinchi marta shu usulda sintez qilingan.

4) Digalogenalkanlardan (degidrogalogenlash)

Digalogenalkanlarning ikki turi bor: **vitsinal** – galogen atomlari ikki qo‘shni uglerod atomida, **geminal** – ikkala galogen atomi bitta uglerod atomida joylashgan. Ularga ishqorning spirtli eritmasi (yoki natriy amid $NaNH_2$) ta’sir ettirilganda 2 molekula vodorod galogenid ajralib, alkin hosil bo‘ladi:

$$CH_2Br–CH_2Br + 2KOH \xrightarrow{\text{spirt, t}} HC\equiv CH + 2KBr + 2H_2O$$

$$CH_3–CH_2–CBr_2–CH_3 + 2KOH \xrightarrow{\text{spirt, t}} CH_3–C\equiv C–CH_3 + 2KBr + 2H_2O$$

Vitsinal digalogenalkanlar alkenlarga galogen biriktirib ($CH_2=CH_2 + Br_2 \rightarrow CH_2Br–CH_2Br$), geminal digalogenalkanlar esa aldegid yoki ketonlarga PCl_5 ta’sir ettirib olinadi ($CH_3–CO–CH_3 + PCl_5 \rightarrow CH_3–CCl_2–CH_3 + POCl_3$).

5) Tetragalogenalkanlarga rux ta’sir ettirish

$$CHBr_2–CHBr_2 + 2Zn \longrightarrow HC\equiv CH + 2ZnBr_2$$

6) Atsetilen gomologlarini olish (atsetilenidlarni alkillash)

Avval natriy atsetilenid olinadi, so‘ngra unga galogenalkan ta’sir ettiriladi – uglerod zanjiri uzayadi:

$$HC\equiv CH + NaNH_2 \longrightarrow HC\equiv C–Na + NH_3$$

$$HC\equiv C–Na + CH_3–Br \longrightarrow HC\equiv C–CH_3 + NaBr$$

$$Na–C\equiv C–Na + 2C_2H_5–Br \longrightarrow C_2H_5–C\equiv C–C_2H_5 + 2NaBr$$ geksin-3

Laboratoriyada atsetilen kalsiy karbidga suv tomizib olinadi; **sanoatda** asosan metan pirolizi va karbid usulidan foydalaniladi.

5. Kimyoviy xossalari

Ikkita π-bog‘ borligi uchun alkinlar alkenlarga o‘xshab **birikish**, **oksidlanish** va **polimerlanish** reaksiyalariga kirishadi. Birikish ikki bosqichda boradi: avval bitta π-bog‘ uzilib alken hosilasi, so‘ngra ikkinchi π-bog‘ uzilib to‘yingan birikma hosil bo‘ladi. Bundan tashqari, uchbog‘i zanjir oxirida joylashgan alkinlarga vodorodning metallga almashinishi – **o‘rin olish** reaksiyalari xos.

I. Birikish reaksiyalari

1) Gidrogenlash (Ni, Pt yoki Pd katalizatori)

$$HC{\equiv}CH + H_2 \xrightarrow{Pd} CH_2{=}CH_2$$

$$CH_2{=}CH_2 + H_2 \xrightarrow{Ni,\ t} CH_3{-}CH_3$$

$$HC{\equiv}C{-}CH_3 + 2H_2 \xrightarrow{Ni,\ t} CH_3{-}CH_2{-}CH_3$$

Faolligi pasaytirilgan («zaharlangan») palladiy katalizatorida reaksiyani alken bosqichida to‘xtatish mumkin.

2) Galogenlash

$$HC{\equiv}CH + Br_2 \longrightarrow CHBr{=}CHBr$$ 1,2-dibrometen

$$CHBr{=}CHBr + Br_2 \longrightarrow CHBr_2{-}CHBr_2$$ 1,1,2,2-tetrabrometan

$$HC{\equiv}CH + 2Cl_2 \longrightarrow CHCl_2{-}CHCl_2$$ 1,1,2,2-tetraxloretan

Bromli suv rangsizlanadi – to‘yinmagan birikmalarga, jumladan alkinlarga sifat reaksiya. Galogenlarning alkinlarga birikishi alkenlarga birikishiga nisbatan sekinroq boradi.

3) Galogenvodorodlarning birikishi

Simmetrik bo‘lmagan alkinlarda **Markovnikov qoidasi** bajariladi: vodorod atomi vodorodi ko‘proq bo‘lgan uglerodga, galogen atomi esa vodorodi kam bo‘lgan uglerodga birikadi. Ikkinchi molekula galogenvodoroddagi galogen atomi ham birinchi galogen turgan uglerodga birikadi:

$$HC{\equiv}CH + HCl \xrightarrow{HgCl_2,\ t} CH_2{=}CHCl$$ vinilxlorid

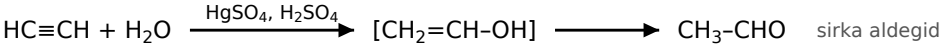
$$CH_2{=}CHCl + HCl \longrightarrow CH_3{-}CHCl_2$$ 1,1-dixloretan

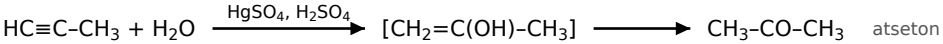
$$HC{\equiv}C{-}CH_3 + HBr \longrightarrow CH_2{=}CBr{-}CH_3$$ 2-brompropen

$$CH_2{=}CBr{-}CH_3 + HBr \longrightarrow CH_3{-}CBr_2{-}CH_3$$ 2,2-dibrompropan

4) Gidratlanish – Kucherov reaksiyasi (M.G. Kucherov, 1881-yil)

Simob(II) tuzlari ($HgSO_4$) ishtirokida, kislotali muhitda alkinlarga suv birikadi. Avval beqaror enol (–OH guruhi qo‘shbog‘dagi uglerodga bog‘langan birikma) hosil bo‘ladi va u darhol karbonil birikmaga izomerlanadi (Eltekov qoidasi):



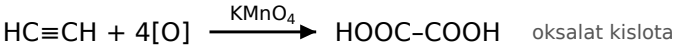


Kucherov reaksiyasida **faqat atsetilendan aldegid** (sirka aldegid), uning gomologlaridan esa **ketonlar** hosil bo‘ladi.

II. Oksidlanish

1) Kaliy permanganat ta’siri (yumshoq oksidlanish)

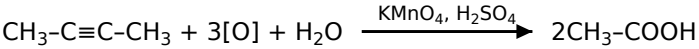
Alkinlar $KMnO_4$ eritmasini **rangsizlantiradi** (sifat reaksiya). Neytral muhitda atsetilen oksalat kislotagacha (uning tuzigacha) oksidlanadi:

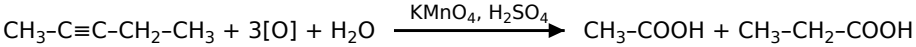


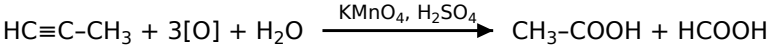
$3C_2H_2 + 8KMnO_4 \rightarrow 3K_2C_2O_4 + 8MnO_2\downarrow + 2KOH + 2H_2O$

2) Qattiq oksidlanish ($KMnO_4 + H_2SO_4$)

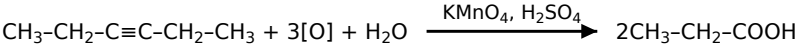
Kislotali muhitda molekula uchbog‘ turgan joydan uziladi va ikki molekula karbon kislota hosil bo‘ladi: $R_1–C\equiv C–R_2 \rightarrow R_1–COOH + R_2–COOH$. Uchbog‘ zanjir oxirida bo‘lsa, ≡CH qismi chumoli kislota (HCOOH) ga aylanadi.







$5CH_3–C\equiv CH + 6KMnO_4 + 9H_2SO_4 \rightarrow 5CH_3COOH + 5HCOOH + 6MnSO_4 + 3K_2SO_4 + 4H_2O$



$5C_2H_5–C\equiv C–C_2H_5 + 6KMnO_4 + 9H_2SO_4 \rightarrow 10C_2H_5COOH + 6MnSO_4 + 3K_2SO_4 + 4H_2O$

3) Yonish

Atsetilen havoda tutab, yorug‘ alanga bilan yonadi, chunki molekulasida uglerodning massa ulushi katta. Kislorodda yonganda alanga harorati 3000 °C dan oshadi:

$$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$$

$$C_nH_{2n-2} + (1{,}5n - 0{,}5)O_2 \rightarrow nCO_2 + (n-1)H_2O$$

III. O‘rin olish – atsetilenidlar hosil bo‘lishi

Uchbog‘i zanjir oxirida joylashgan alkinlarning ≡C–H vodorodi metall atomlariga almashinadi va **atsetilenidlar** (metall karbidlari) hosil bo‘ladi:

$$HC{\equiv}CH + 2Na \xrightarrow{t} Na{-}C{\equiv}C{-}Na + H_2$$ natriy atsetilenid

$$HC{\equiv}CH + Ag_2O \xrightarrow{NH_3\ \text{eritmasi}} Ag{-}C{\equiv}C{-}Ag{\downarrow} + H_2O$$ kumush atsetilenid

To‘liq tenglama: $HC{\equiv}CH + 2[Ag(NH_3)_2]OH \rightarrow Ag{-}C{\equiv}C{-}Ag{\downarrow} + 4NH_3 + 2H_2O$ (oq cho‘kma).

$$HC{\equiv}CH + 2[Cu(NH_3)_2]Cl \longrightarrow Cu{-}C{\equiv}C{-}Cu{\downarrow} + 2NH_4Cl + 2NH_3$$

Mis(I) atsetilenid – qizil-qo‘ng‘ir cho‘kma.

$$HC{\equiv}C{-}CH_3 + [Ag(NH_3)_2]OH \longrightarrow Ag{-}C{\equiv}C{-}CH_3{\downarrow} + 2NH_3 + H_2O$$

$$CH_3{-}C{\equiv}C{-}CH_3 + [Ag(NH_3)_2]OH \longrightarrow \text{reaksiya bormaydi}$$

Sifat reaksiya: kumush oksidning ammiakli eritmasi bilan oq, mis(I) xloridning ammiakli eritmasi bilan qizil-qo‘ng‘ir cho‘kma hosil bo‘lishi uchbog‘i zanjir oxirida joylashgan alkinlarni (masalan, butin-1 ni) butin-2 kabi alkinlardan farqlash imkonini beradi.

Atsetilenidlarga kislota ta’sir ettirilsa, qaytadan alkin ajralib chiqadi:

$$Ag{-}C{\equiv}C{-}Ag + 2HCl \longrightarrow HC{\equiv}CH{\uparrow} + 2AgCl{\downarrow}$$

Ehtiyot bo‘ling: og‘ir metallarning atsetilenidlari (Ag_2C_2, Cu_2C_2) quruq holatda zarbadan yoki qizdirilganda portlaydi, shuning uchun ular faqat nam holatda saqlanadi.

IV. Polimerlanish

1) Dimerlanish

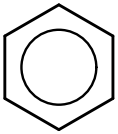
Mis(I) xlorid va ammoniy xlorid ishtirokida ikki molekula atsetilen birikib, vinilatsetilen hosil qiladi:

$$2HC{\equiv}CH \xrightarrow{CuCl,\ NH_4Cl} CH_2{=}CH{-}C{\equiv}CH$$ vinilatsetilen

Vinilatsetilenga HCl biriktirib xloropren – xloropren kauchugining monomeri olinadi: $CH_2{=}CH{-}C{\equiv}CH + HCl \rightarrow CH_2{=}CCl{-}CH{=}CH_2$.

2) Trimerlanish (N.D. Zelinskiy, B.A. Kazanskiy)

Atsetilen aktivlangan ko‘mir ustidan qizdirib o‘tkazilganda uch molekulasi birikib, benzol hosil qiladi:

$$3HC{\equiv}CH \xrightarrow{C\ (aktiv.),\ 450–500\ °C}$$  benzol

6. Ishlatilishi

Atsetilen – organik sintezning muhim xomashyosi va yuqori haroratli alanga beruvchi yoqilg‘i.

- **Metallarni payvandlash va kesish:** atsetilen kislorodda yonganda alanga harorati 3000 °C dan oshadi (avtogen payvandlash).
- **Vinilxlorid** → polivinilxlorid (quvurlar, linoleum, elektr izolyatsiyasi, sun’iy charm).
- **Sirka aldegid** (Kucherov reaksiyasi) → sirka kislota, etil spirt va boshqa moddalar.
- **Vinilatsetilen** → xloropren → xloropren kauchugi (neopren).
- **Erituvchilar:** 1,1,2,2-tetraxloretan $CHCl_2{-}CHCl_2$ va trixloretilen yog‘lar, smolalar va ko‘plab organik moddalarni yaxshi eritadi; ular yonmaydi, ya’ni o‘t olish xavfi yo‘q.
- **Benzol** (trimerlanish) va boshqa aromatik birikmalar olishda.

Mahsulot	Olinishi	Ishlatilishi
vinilxlorid	$HC{\equiv}CH + HCl\ (HgCl_2)$	polivinilxlorid: quvur, linoleum
sirka aldegid	$HC{\equiv}CH + H_2O\ (HgSO_4)$	sirka kislota olish
vinilatsetilen	$2HC{\equiv}CH\ (CuCl,\ NH_4Cl)$	xloropren kauchugi
benzol	$3HC{\equiv}CH$ (C, t)	organik sintez

Saqlanishi: siqilgan atsetilen o‘z-o‘zidan parchalanib portlashi mumkin, shuning uchun u po‘lat ballonlarda g‘ovak massaga shimdirilgan atsetonda eritilgan holda saqlanadi.

O‘zingni tekshir

1. C_5H_8 tarkibli alkinlarning barcha izomerlarini yozing va nomlang.

2. 4-metilpentin-2 ning ratsional nomini ayting.

3. Nima uchun alkinlarda sis-trans izomeriya bo‘lmaydi?

4. Propinga ketma-ket 2 molekula HBr birikkanda qanday modda hosil bo‘ladi?

5. Kucherov reaksiyasi bo‘yicha atsetilen va propindan qanday moddalar hosil bo‘ladi?

6. Butin-1 va butin-2 ni qaysi reaktiv yordamida farqlash mumkin?

7. 6,4 g kalsiy karbiddan necha litr (n.sh.) atsetilen olish mumkin?

8. 2,6 g atsetilen bilan necha gramm brom to‘liq birikadi?

9. 5,6 l (n.sh.) atsetilen to‘liq yonganda necha litr (n.sh.) CO_2 hosil bo‘ladi?

Javoblar: 1) 3 ta: pentin-1, pentin-2, 3-metilbutin-1; 2) metilizopropilatsetilen; 3) uchbog‘dagi uglerodlar sp-gibridlangan, har birida bittadan o‘rinbosar bor va molekulaning bu qismi chiziqli; 4) 2,2-dibrompropan (Markovnikov qoidasi); 5) sirka aldegid va atseton; 6) kumush oksidning ammiakli eritmasi: butin-1 cho‘kma hosil qiladi, butin-2 – yo‘q; 7) $n(CaC_2) = 0{,}1$ mol, $V(C_2H_2) = 2{,}24$ l; 8) $n(C_2H_2) = 0{,}1$ mol, $n(Br_2) = 0{,}2$ mol, $m(Br_2) = 32$ g; 9) $n(C_2H_2) = 0{,}25$ mol, $V(CO_2) = 11{,}2$ l.